Inspire the love of learning

WOG

Q1. MAP vs. ML estimation

Given a coin with prior $P(\theta=0.6)=0.7$ and $P(\theta=0.5)=0.3$

Observe $D=\{H,H,T\}$

a) Compute the likelihood $P(D|\theta)$ for each hypothesis

b) Find the MAP estimate $\hat{\theta}_{MAP}$

Ans:

a)

$P(D|\theta=0.6)=0.6^2\times 0.4=0.144$

$P(D|\theta=0.5)=0.5^3=0.125$

b) Posterior $\propto$ likelihood $\times$ prior:

$P(\theta=0.6|D)\propto 0.144\times 0.7=0.1008$

$P(\theta=0.5|D)\propto 0.125\times 0.3=0.0375$

$\hat{\theta}_{MAP}=0.6$ (higher unnormalised posterior)

ML would also choose 0.6 (higher likelihood) but MAP and ML can disagree when the prior is strong enough to override

Q2. Naive Bayes Classification

Two classes c_1, c_2 with $P(c_1)=P(c_2)=0.5$. Two binary features:

	$P(x_1\|c)$		$P(x_2\|c)$	
c_1	0.3	0.7	0.8	0.2
c_2	0.6	0.4	0.4	0.6

Classify $x=(1,1)$

$P(c_1|x)\propto P(c_1)\times P(x_1=1|c_1)\times P(x_2=1|c_1)$

$=0.5\times 0.7\times 0.2=0.07$

$P(c_2|x)\propto P(c_2)\times P(x_1=1|c_2)\times P(x_2=1|c_2)$

$=0.5\times 0.4\times 0.6=0.12$

Normalize: $P(c_1|x)=\frac{0.07}{0.07+0.12}\approx 0.368$

$P(c_2|x)=\left(\frac{0.12}{0.07+0.12}\right)\approx 0.632$

$\Rightarrow$ Predict $\hat{y}=c_2$ (63,2%)

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Q3 Gaussian Naive Bayes

A data set has 2 features. Class A: $\mu = [0,0]$, $\sigma^2 = [4,0]$ (feature 2 is const). Explain why zero variance is problematic & how var_smoothing fix it.

With $\sigma^2 = 0$ the Gaussian pdf $\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ involves division by zero. Smoothing add ϵ = var_smoothing × max$(\sigma^2) = 10^{-9} \times 4 = 4.10^{-9}$ to all variances, giving $\sigma^2 = 4.10^{-9} > 0$. This keeps the pdf well-defined while barely affecting features with large variance. The ϵ scales with data magnitude rather than being a fixed constant

Q4 Conditional Independence Assumption

State the conditional independence assumption of Naive Bayes. Give 1 example where it is clearly violated and explain why NB can still perform well despite this violation.

- Assumption: $P(x_1, \ldots, x_d | Y=c) = \prod_{j=1}^{d} (P x_j | Y=c)$
- Violated example: In text classification, the word "machine" and "learning" co-occur far more often than independence predicts: P("machine", "learning" | c) >> P("machine" | c) × P("learning" | c).
- Why NB still work: Classification only requires the correct ranking of posterior probabilities, not their calibrated values. Even if P(c|x) is poorly estimated in absolute terms, the argmax can still be correct. Domingos & Pazzani (1997) showed NB is "an amazingly good classifier" because the decision boundary is robust to the independene violation.

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